

PROJECT REPORT: MOVIE RECOMMENDATION SYSTEM

Content-Based Machine Learning & Relational Analytics

Author: Om Vadiya | **Career Focus:** Data Science & Machine Learning
Degree: Bachelor of Commerce (Graduation: July 2026, CGPA: 7.08/10)
Institution: M J College of Commerce, Maharaja Krishnakumarsinhji Bhavnagar University
Profiles: GitHub: github.com/vadiyaom | LinkedIn: linkedin.com/in/Vadiya-Om/ | Email: vadiyaom18@gmail.com

1. Executive Summary & Introduction

With the rapid expansion of digital media catalogs, consumers encounter significant cognitive load when selecting content. This project develops an end-to-end, production-grade Content-Based Movie Recommendation System. By leveraging Natural Language Processing (NLP), TF-IDF vectorization, Cosine Similarity, and relational SQL analytics, the application maps unstructured movie metadata into high-dimensional vector spaces to surface top-N thematic matches with zero runtime latency.

2. Problem Statement & Objectives

Standard filtering by year or broad category fails to capture intricate narrative affinities such as directorial nuances, thematic motifs, and ensemble chemistry. The core objective is to calculate pairwise movie affinities using intrinsic metadata without relying on user rating histories or invasive tracking, making it privacy-preserving and highly explainable.

3. Technologies & Architecture

Technology	Role / Purpose
Python 3.10+	Core programming language and pipeline orchestration
Pandas & NumPy	Data preprocessing, dataframe operations, vector mathematics
Scikit-Learn	TF-IDF Vectorization and Cosine Similarity calculation
SQL (SQLite)	Relational schema design, catalog aggregations, CTEs, subqueries
Streamlit	Interactive web application UI and model serving
Git & GitHub	Version control, repository structure, portfolio presentation

4. Machine Learning Methodology

- Feature Engineering:** Five descriptive text attributes (genres, keywords, overview, cast, director) are cleaned, lowercased, and concatenated into a unified text document per movie.
- TF-IDF Vectorization:** The corpus is transformed into a weighted sparse matrix using unigram and bigram tokens (`ngram_range=(1,2)`), penalizing ubiquitous stop words and magnifying discriminative terms.
- Cosine Similarity:** Evaluates the cosine of the geometric angle between normalized vectors. Cosine similarity is uniquely suited because it neutralizes differences in document lengths and computes thematic directional alignment.

- **Performance Optimization:** Similarity matrices are precomputed at startup, ensuring sub-10ms query execution in production.

5. Relational SQL Analytics Showcase

The project repository features a comprehensive SQL suite (*sql/analysis_queries.sql*) demonstrating:

- **Filtering & Aggregation:** Summary statistics, AVG ratings, decade distributions.
- **GROUP BY & HAVING:** Director performance analytics filtering creators with minimum 2 catalog entries.
- **Multi-Table Joins:** Many-to-many joins across movies, movie_genres, and genres lookup tables.
- **Subqueries & CTEs:** Scalar subqueries identifying above-average films and window ranking (DENSE_RANK) across genres.

6. Project Results & Interview Readiness

Testing against benchmark titles (e.g., *The Dark Knight* returning *Batman Begins* and *The Prestige*; *Inception* returning *Tenet* and *Interstellar*) validates robust semantic affinity. This project provides an authentic, comprehensive showcase of Python, Data Science workflows, Machine Learning fundamentals, SQL fluency, and web application deployment.